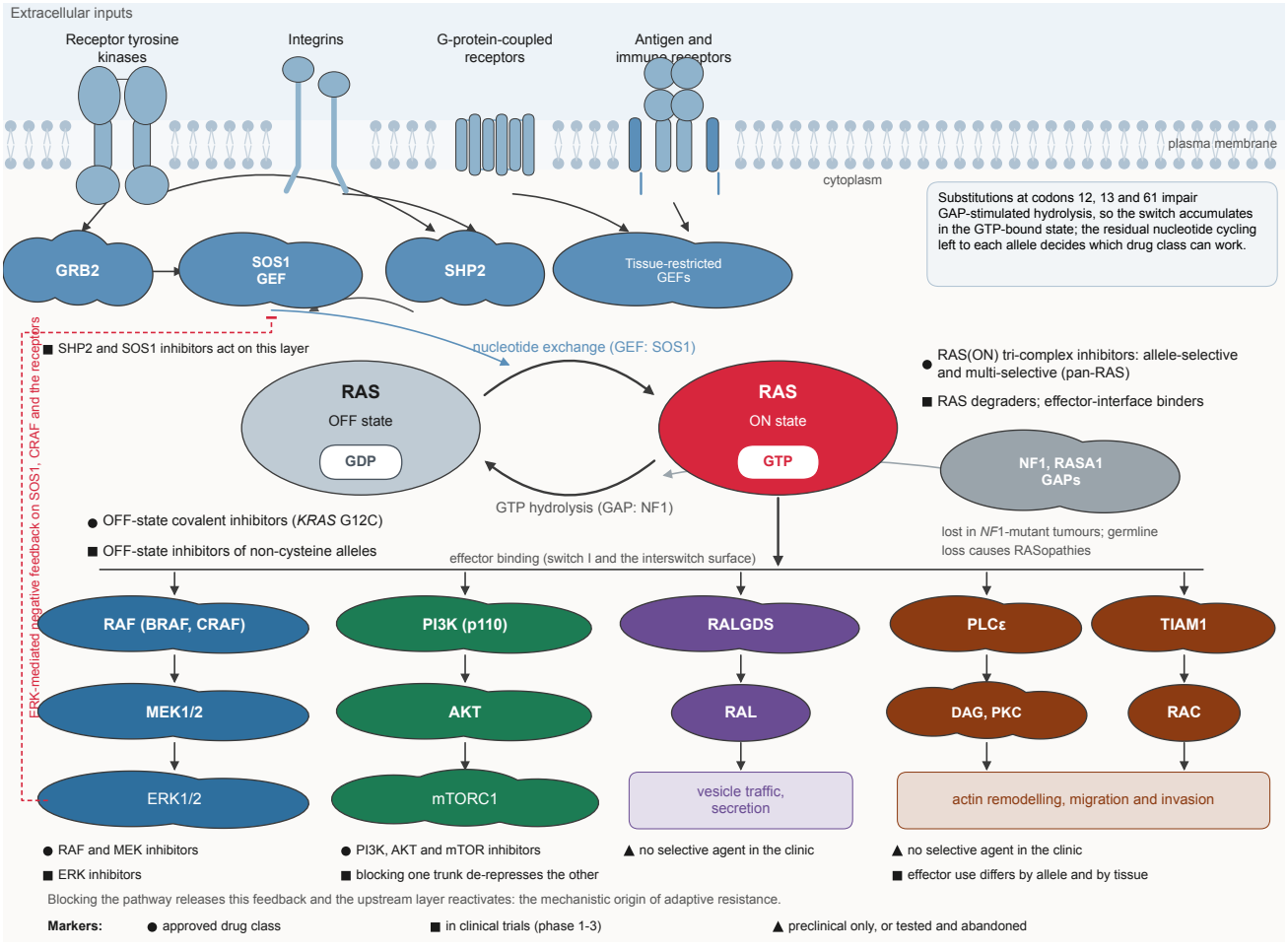
a




b

Cellular output of sustained RAS signalling (review section 3.5)

**Metabolic reprogramming**

Increased glucose uptake, rewired glutamine use, scavenging of extracellular nutrients

▲ metabolic co-targeting

**Macropinocytosis and autophagy**

Protein and amino-acid recycling sustains growth and becomes a survival route under drug pressure

■ autophagy inhibition

**Immunosuppressive microenvironment**

Higher PD-L1, recruitment of myeloid suppressor cells, reduced antigen presentation

■ checkpoint blockade; mutant-RAS vaccines and TCR-T cells

**Senescence and oncogenic stress**

A barrier that must be bypassed: carrying a mutation is not the same as depending on it

▲ no therapeutic use; constrains how mutation data are read